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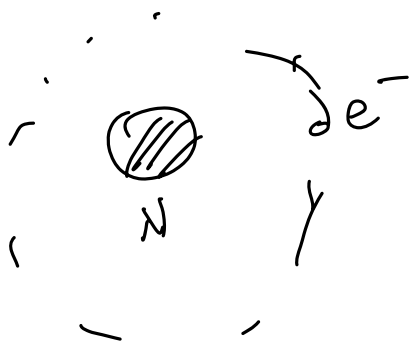
radiation.net



Classroom 2026 Particle Physics



Bachelor's course:



Rutherford experiment

$$\alpha + N \rightarrow \alpha + N$$

$$K_{\alpha} \approx 5 \text{ MeV}$$

In this course $E \gg \text{MeV}$

$$\underline{P} = (E, \vec{P})$$

$$E^2 = \vec{P}^2 + m^2$$

$$\vec{P}^2 = |\vec{P}|^2$$

Natural units: $\hbar = c = 1$ dimensionless

$$E = h\nu = \hbar\omega$$

$$[\hbar] = \frac{[E]}{[\nu]} = \frac{[E]}{\frac{1}{[T]}} \quad \hbar = \frac{h}{2\pi}$$

$$\hbar \text{ dimensionless} \Rightarrow [E] = [T]^{-1}$$

$$c = 1 \quad [\nu] = \frac{[L]}{[T]}$$

$$c \text{ dimensionless} \Rightarrow [L] = [T]$$

eV for E

$$c = 3 \times 10^8 \text{ m/s}$$

eV⁻¹ for L, T

$$\hbar = 1.05 \times 10^{-34} \text{ J/s}$$

$$\begin{aligned} \hbar c &= 200 \text{ MeV} \times \text{fm} \\ &= 197 \text{ MeV} \times \text{fm} \end{aligned}$$

$$1 \text{ fm} = 10^{-15} \text{ m} \quad \text{Fermi:}$$

$$1 \text{ MeV} = 10^6 \text{ eV}$$

$$1 \text{ eV} = 1.6 \times 10^{-16} \text{ C} \cdot \text{V} = 1.6 \times 10^{-19} \text{ J}$$

1) Decays

2) Collisions

$$\text{Decay:} \quad a \rightarrow b + c + d + \dots + z$$

neutrons, protons, electrons

neutrons unstable:

protons, electrons: stable.

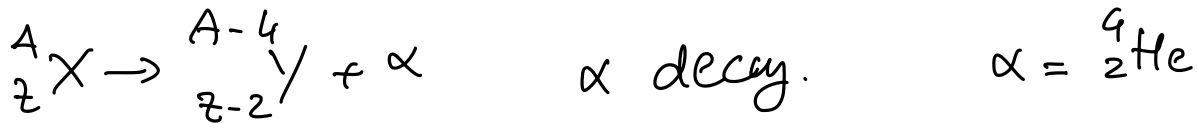
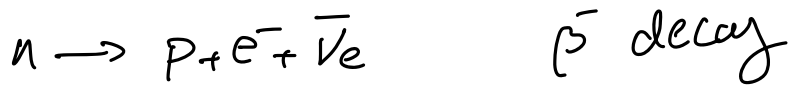
Decay caused by existence of fundamental interactions

$$\begin{aligned} a &\rightarrow b + c \\ &\hookrightarrow \text{interaction} \end{aligned}$$

gravity too weak at subnuclear scale

EM
weak
strong } can cause decay of particles

Energy - Momentum conservation always in all decays



baryons + electrons: visible matter

bosons { baryons: neutrons, protons } strong interaction
mesons: π, K

leptons: $e^-, \mu^-, \bar{\nu}_e, \nu_\mu$ } no strong interaction
only weak and EM

Standard Model of Elementary Particles

three generations of matter (fermions)						interactions / force carriers (bosons)	
I		II		III			
mass charge spin	$\approx 2.16 \text{ MeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$	$\approx 1.273 \text{ GeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$	$\approx 172.57 \text{ GeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$	0 0 1	$\approx 125.2 \text{ GeV}/c^2$ 0 0		
	u up	c charm	t top	g gluon	H higgs		
	$\approx 4.7 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$	$\approx 93.5 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$	$\approx 4.183 \text{ GeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$	0 0 1			
	d down	s strange	b bottom	γ photon			
	$\approx 0.511 \text{ MeV}/c^2$ -1 $\frac{1}{2}$	$\approx 105.66 \text{ MeV}/c^2$ -1 $\frac{1}{2}$	$\approx 1.77693 \text{ GeV}/c^2$ -1 $\frac{1}{2}$	$\approx 91.188 \text{ GeV}/c^2$ 0 1			
	e electron	μ muon	τ tau	Z Z boson			
	$< 0.8 \text{ eV}/c^2$ 0 $\frac{1}{2}$	$< 0.17 \text{ MeV}/c^2$ 0 $\frac{1}{2}$	$< 18.2 \text{ MeV}/c^2$ 0 $\frac{1}{2}$	$\approx 80.3692 \text{ GeV}/c^2$ 0 ± 1			
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	W W boson			

QUARKS

LEPTONS

SCALAR BOSONS

GAUGE BOSONS
VECTOR BOSONS

baryons: bound state of 3 quarks $q_1 q_2 q_3$

: uud

$$Q = +\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +\frac{2}{3} = +1 \quad \text{proton.}$$

udd:

$$Q = \frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0 \quad \text{neutron.}$$

Mesons: $q_i \bar{q}_j$

pions: $\pi^+ = u\bar{d}$

$\bar{d}$: anti particle of d quark anti-d

$$Q_{\bar{d}} = -Q_d$$

$$Q : d = -1/3$$

$$Q : \bar{d} = +1/3$$

$$Q_u = +\frac{2}{3}$$

$$Q_{\bar{u}} = -\frac{2}{3}$$

leptons
quarks: Fundamental / elementary particles

baryons, mesons (hadrons) : are not elementary
bound / Composite states

Elementary particle?

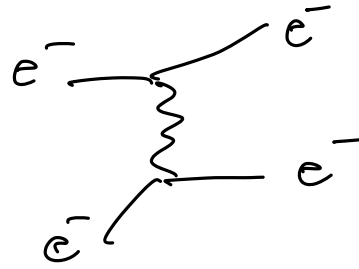
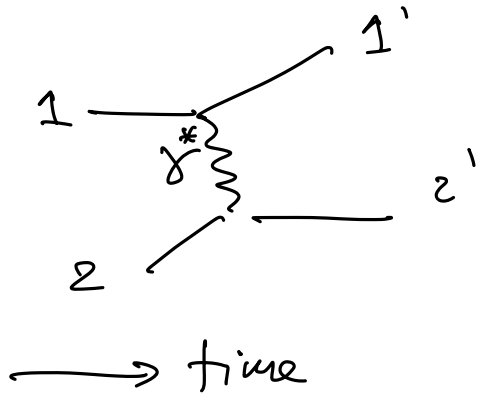
all transferred energy becomes kinetic energy
for the particle

$$\begin{array}{c} \text{~~~~~} \rightarrow \bullet \\ \Delta E \end{array} \quad K_i = 0.$$

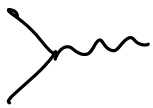
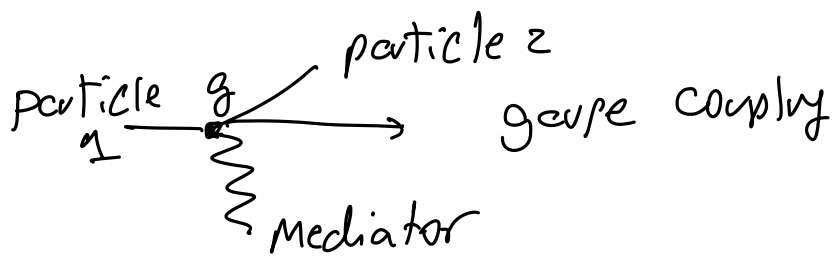
$$K_f = 0 + \Delta E$$

Elementary particles do not have internal degrees of freedom (think about rotational energy of a rigid body)

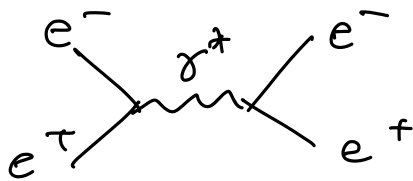
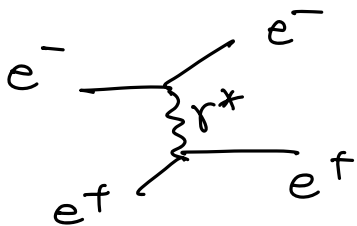
Mediators of forces for gauge interactions:



$e^- + e^- \rightarrow e^- + e^-$
Möller scattering



Interaction
vertices



Decays

$$a \rightarrow b + c + \dots + d$$

N-particle decay.

$$a \rightarrow b + c$$

2-body decay

⋮

$$a \rightarrow z + w + x$$

Sample of
unstable particles

$$t=0$$

$N = N_0$ particles

Γ : prob. of decay per unit time

$$N(t) = N(0) e^{-\Gamma t}$$

particles @ time t
surviving

$$\frac{N(t)}{N(0)} = e^{-\Gamma t} \leq 1$$

probability of survival.

$$[\Gamma] = T^{-1} = [E] \quad \text{in natural units.}$$

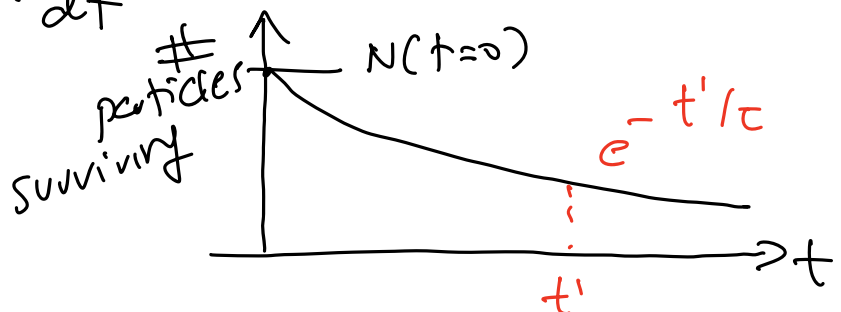
lifetime: property of unstable particles

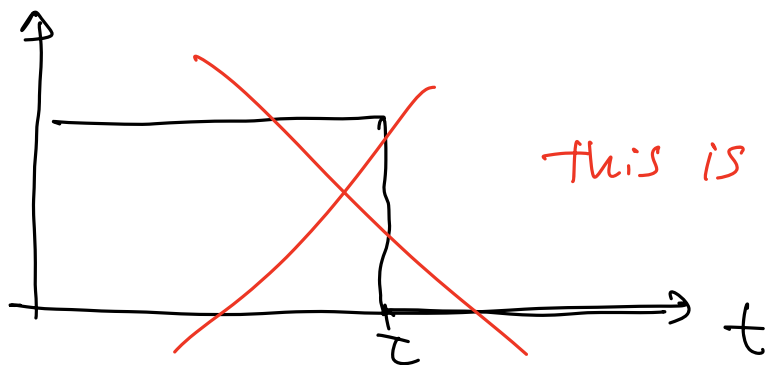
average life of these unstable particles.

has to take into account decay prob. (weighted average)

$$\tau = \langle t \rangle = \frac{\int_0^{\infty} t e^{-\Gamma t} dt}{\int_0^{\infty} e^{-\Gamma t} dt} = \frac{1}{\Gamma}$$

$$N(t) = N_0 e^{-t/\tau}$$



$N(t)$ 

this is not lifetime

μ ons: $\tau = 2.2 \times 10^{-6}$ sec.

τ : is in reference frame where particle at rest

in the lab the effective lifetime will be longer

LAB

$\beta = \frac{v}{c}$ $\gamma = \frac{1}{\sqrt{1-\beta^2}}$

}

$$t = \gamma(t' + \beta z')$$

$$x = x'$$

$$y = y'$$

$$z = \gamma(z' + \frac{v}{c}t')$$

$$= \gamma(\beta t' + z')$$

$$\tau_{LAB} = \gamma \tau$$

$$\beta = \frac{|\vec{p}|}{E} \quad \gamma = \frac{E}{m} \quad \beta\gamma = \frac{|\vec{p}|}{m}$$

$$a \rightarrow b + c$$

Energy and mom. must be conserved.

$\Rightarrow$ Q value of decay.

$$E_a = E_b + E_c \quad \text{en. conservation.} \quad E = \sqrt{p^2 + m^2}$$

In ref with $\vec{P}_a = 0 \Rightarrow E_a = M_a$

$$E = m + \underbrace{K}_{\text{definition of kinetic energy.}}$$

$$c \leftarrow \overset{a}{\bullet} \longrightarrow b$$

$$\vec{0} = \vec{P}_b + \vec{P}_c \quad \text{mom. cons.} \Rightarrow \vec{P}_b = -\vec{P}_c$$

$$M_a = E_b + E_c = m_b + K_b + m_c + K_c = (m_b + m_c) + (K_b + K_c)$$

$$Q = M_a - m_b - m_c = K_b + K_c \geq 0$$

Decays are possible only if $Q \geq 0$

$$a \longrightarrow b + c + \dots + d + \dots + z$$

$$Q = M_a - m_b - m_c - m_d - \dots - m_z \geq 0$$

Q small $\rightarrow$ small prob. for decay